

Master Thesis Supervisor Meeting – Authoritative Numbers Cheat Sheet

Mode-I Phase-Field Convergence & UEL Energy Audit · Date: 08 October 2026, 10:00

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1. Authoritative Mechanical Benchmark Numbers (Job Provenance Audited)

Simulation Model	Elements	K_0 [kN/mm]	$F_{\max}$ [kN]	$u(F_{\max})$ [mm]	Status & Extraction Provenance
Fixed Reference Anchor (1398090.mm _{master02})	15,192 (CPE4)	137.945520 (Ref Anchor)	0.757778 (Ref Anchor)	0.005857 (Ref Anchor)	Unconstrained OLS, $N = 400$ ($u \leq 1.0\text{ }\mu\text{m}$), $R^2 = 0.99999960$, $b = 4.472 \times 10^{-5}\text{ kN}$
Corrected Nominal 1% (1404933.mm _{master02})	71,320 (69kQ + 1.8kT)	137.820804 (−0.0904%)	0.745325 (−1.64%)	0.005750 (−1.83%)	Authoritative full-fracture solve: $K_0 \approx 137.821$, $N = 400$, $R^2 = 0.99999960$, $b = 4.470 \times 10^{-5}\text{ kN}$
Spatial Fine S_3 ($h = 1.5\text{ }\mu\text{m}$) (1406017.mm _{master02})	41,912 (CPE4)	137.857612 (−0.0637%)	0.732185 (−3.38%)	0.005633 (−3.82%)	Spatial mesh sensitivity anchor ($h/l_0 = 0.200$); cutback at $u = 7.84\text{ }\mu\text{m}$
Temporal Fine T_3 ($0.5 \times \Delta t$) (1406317.mm _{master02})	15,192 (CPE4)	137.945892 (+0.0003%)	0.757634 (−0.0190%)	0.005855 (−0.0341%)	Authoritative full horizon to $u = 10.0\text{ }\mu\text{m}$; 10,022 Step-2 increments, <code>Exit_status=0</code>
Diagnostic Twin (1406839.mm _{master02})	15,192 (CPE4)	137.945520 (0.0000%)	0.757778 (0.0000%)	0.005857 (0.0000%)	Full diagnostic twin: exact bit-for-bit parity; 7,000 incs, terminal $\Delta_{\text{book}} = +0.7607\%$

2. UEL Energy Bookkeeping & Parity Numbers

A. Terminal Global Energy Bookkeeping (S_1 , $u = 10.0\text{ }\mu\text{m}$):

- External Work $W_{\text{ext}} = W_{\text{trap}}$: **$2.35933 \times 10^{-3}\text{ kN} \cdot \text{mm}$** (2.35933 mJ).
- Elastic Strain Energy E_{elas} : **$1.16081 \times 10^{-6}\text{ kN} \cdot \text{mm}$** (0.00116 mJ, residual).
- Fracture Surface Energy E_{frac} : **$2.34022 \times 10^{-3}\text{ kN} \cdot \text{mm}$** (99.19% of work).
- Two-Term Bookkeeping Difference Δ_{book} : **$1.795 \times 10^{-5}\text{ kN} \cdot \text{mm}$** (+**0.7607%**).

B. Non-Invasive Mechanical Parity Verification:

- Elastic Stage (Jobs 1406904 vs 1406905): $\max|\Delta F| = \mathbf{0.00000000\text{ kN}}$ (30 incs).
- Coupled Fracture (Jobs 1406906 vs 1406907): $\max|\Delta F| = \mathbf{0.0\text{ kN}}$ (129 incs to $u = 35\text{ }\mu\text{m}$).
- Single-IP Extraction Rule: eliminates $4\times$ overcounting on CPE4 visualizers.

C. Crack Symmetry and Localization Band Metrics:

- Crack Centroid Trajectory: $|y_c - 0.500\text{ mm}| \leq \mathbf{3.10\text{ }\mu\text{m}}$ across all meshes.
- Fixed meshes (S_1 – S_4): $|y_c - 0.5| \leq 3.10\text{ }\mu\text{m}$; adaptive (A_1 – A_4): $\leq 1.50\text{ }\mu\text{m}$.
- Moderate Damage Band ($d \geq 0.50$): $w_{d \geq 0.5} = \mathbf{20.0\text{--}30.0\text{ }\mu\text{m}}$ (resolution-limited).
- Core Separation Zone ($d \geq 0.90$): $w_{d \geq 0.9} = \mathbf{9.5\text{--}14.4\text{ }\mu\text{m}}$.

D. Preserved Adaptive Sensitivity Trends:

- `errorTarget` = 1.0% $\rightarrow$ **71,320** finite elements.
- `errorTarget` = 2.0% $\rightarrow$ **17,687** (offline) / **15,396** (production) elements.
- `errorTarget` = 3.0% $\rightarrow$ **8,120** (offline) / **7,633** (production) elements.
- `errorTarget` = 5.0% $\rightarrow$ **4,356** (offline) / **4,194** (production) elements.

3. Multi-Quantity Convergence & Epistemological Status Summary

Evaluated Quantity	Observed Variation	Authoritative Status	Epistemological Grounding / Evidence Boundary
Initial Stiffness K_0 (Spatial)	$\Delta < 0.089\%$ across S_1 – S_5	CONVERGED / STABLE	Stable linear elasticity ($u \leq 1.0\text{ }\mu\text{m}$); insensitive to crack-tip damage.
Initial Stiffness K_0 (Temporal)	$\Delta < 0.001\%$ across T_1 – T_3	CONVERGED / STABLE	Time incrementation does not perturb linear elastic stiffness.
Peak Reaction Force $F_{\max}$ (Spatial)	$0.7578 \rightarrow 0.7255\text{ kN}$ (4.26%)	MESH-SENSITIVE	Monotonic decrease with h/l_0 ; asymptotic limit requires sub-element models.
Peak Reaction Force $F_{\max}$ (Temporal)	$\Delta < 0.069\%$ across T_1 – T_3	CONVERGED / STABLE	Limit load invariant under time-step refinement at fixed spatial mesh.
Crack Centroid Symmetry $y_c(x)$	$ y_c - 0.5\text{ mm} \leq 3.10\text{ }\mu\text{m}$	CONVERGED / STABLE	Centroid offset strictly bounded within single element dimension.
Two-Term Bookkeeping Δ_{book}	$+0.38\%$ to $+3.54\%$ ($T_1 \rightarrow T_3$)	TEMPORALLY SENSITIVE	Algebraic difference $W_{\text{trap}} - (E_{\text{elas}} + E_{\text{frac}})$; not a proven balance law.
Global Energy Identity	Non-commuting staggered steps	GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED	Standard ODBs unpersisted for internal Newton paths; identity remains open.